CDO Risk Simulation and Waterfall Analysis

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VFE 6103 - Valuation for Financial Engineering | Professor David Shimko
October 5, 2026

Methodology

We modeled a five-year collateralized debt obligation backed by ten identical speculative-grade corporate bonds, each with $10 million face value, a 6% annual coupon paid quarterly, a 4% annual default probability, and 60% loss given default. A constant 0.20 pairwise dependence parameter was imposed through a 10 x 10 Gaussian correlation matrix. Using a fixed random seed, 1,000 sets of independent standard-normal draws were transformed with a Cholesky factor to generate correlated normals. The standard-normal CDF then mapped these draws to correlated uniforms, which were converted to default times using the constant annual default-probability relationship.

Bond cash flows were generated quarterly. Before default, each surviving bond paid its scheduled coupon; in the default quarter it paid recovery equal to 40% of face value and no coupon; after default, no further cash flow was paid. Bonds surviving through quarter 20 paid both the final coupon and face value. Collateral interest was allocated first to Class A interest, then Class B interest, with the residual paid to equity. Recovery and principal proceeds were applied sequentially to Class A principal, then Class B principal, and finally equity. Tranche coupons were calculated on beginning-of-quarter outstanding principal balances. This principal allocation is a modeling assumption because the project description does not specify a detailed amortization schedule.

Table 1. Model Parameters

Input	Base Case
Number of simulations	1,000
Number of bonds	10
Bond face value	$10MM each
Maturity	5 years / 20 quarters
Annual coupon	6% paid quarterly
Annual default probability	4%
LGD / recovery	60% / 40%
Default correlation parameter	0.20
Class A	$20MM notional; 2% coupon
Class B	$10MM notional; 4% coupon

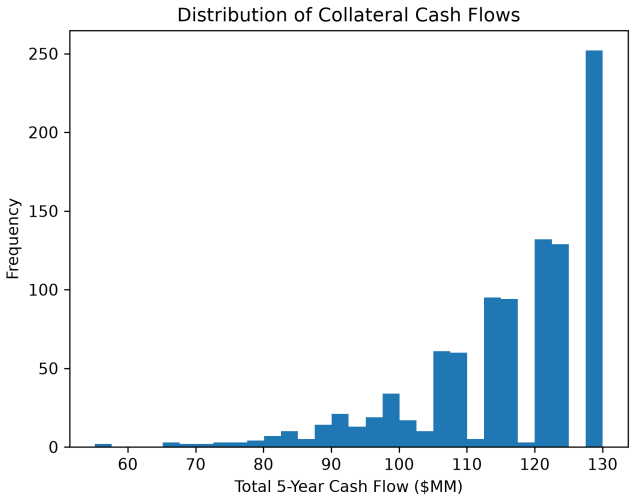
Table 2. Base-Case Five-Year Cash-Flow Results

Component	Mean ($MM)	Std. Dev. ($MM)	5th Pctl. ($MM)	Min ($MM)	Max ($MM)
Collateral	115.94	13.57	89.79	55.15	130.00
Class A	21.65	0.39	20.80	20.16	22.00
Class B	11.99	0.08	12.00	10.76	12.00
Equity	82.31	13.17	57.05	24.23	96.00

Base-Case Cash-Flow Distributions

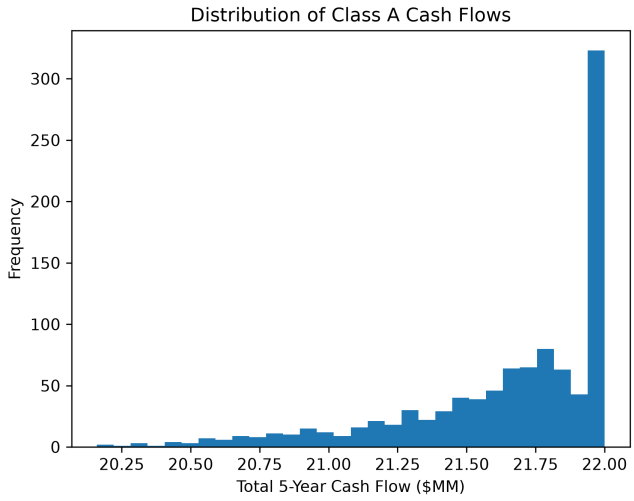
The collateral distribution reflects the effect of correlated defaults on total five-year cash flow. Class A is much more tightly distributed because it receives first priority in the waterfall.

Figure 1. Distribution of Collateral Cash Flows



Total five-year collateral cash flow across 1,000 correlated default scenarios.

Figure 2. Distribution of Class A Cash Flows

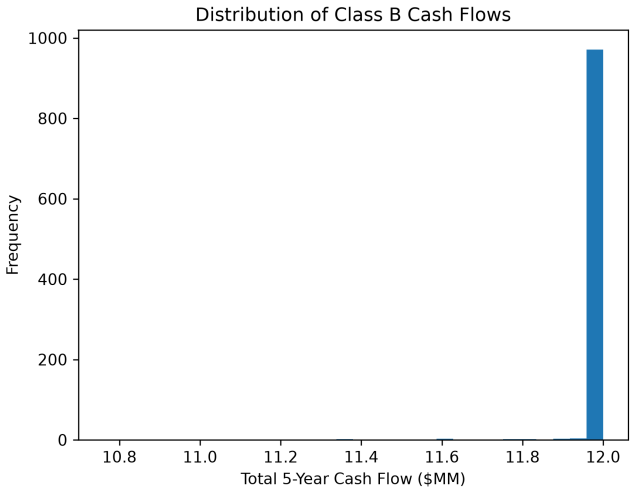


Class A is concentrated near its maximum because of senior payment priority and subordination.

Tranche and Equity Distributions

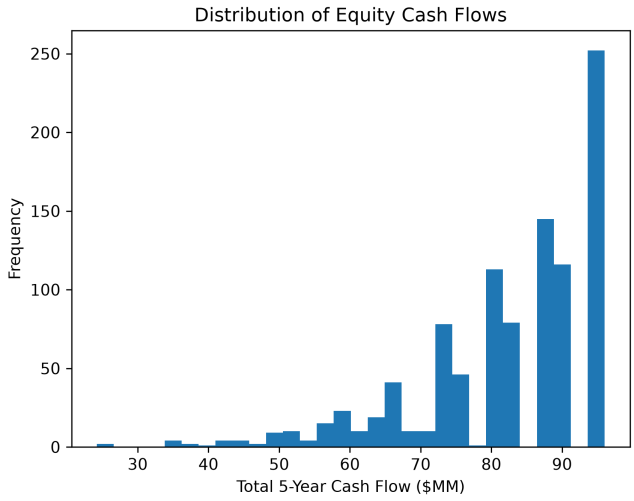
Class B remains strongly protected in the base simulation sample, while equity absorbs most of the variation in collateral performance. The contrast between the two distributions illustrates how subordination transfers credit variability to the residual tranche.

Figure 3. Distribution of Class B Cash Flows



Class B is highly concentrated around $12MM in the base sample.

Figure 4. Distribution of Equity Cash Flows



Equity exhibits the broadest downside distribution because it absorbs residual collateral risk.

Sensitivity Analysis

Sensitivity tests varied default probability, loss given default, and default correlation one at a time while holding the random seed fixed. Higher PD and LGD reduce both expected cash flows and lower-tail outcomes. Correlation has comparatively little effect on the mean but materially changes dispersion and downside risk because it changes the degree to which defaults cluster across obligors.

Table 3. Sensitivity Results (USD millions)

Scenario	Coll. Mean	Coll. Std	Coll. 5%	Eq. Mean	Eq. Std	Eq. 5%
Base	115.94	13.57	89.79	82.31	13.16	57.05
Low PD	122.90	9.49	103.86	89.08	9.23	70.84
High PD	110.03	15.75	80.64	76.57	15.24	48.44
Low LGD	119.64	10.11	99.79	86.21	9.46	68.03
High LGD	112.25	17.03	79.79	78.43	16.82	46.42
Low Corr	116.06	11.62	93.08	82.42	11.29	59.90
High Corr	116.21	16.78	80.63	82.58	16.24	48.30

Figure 5. Collateral Tail Sensitivity

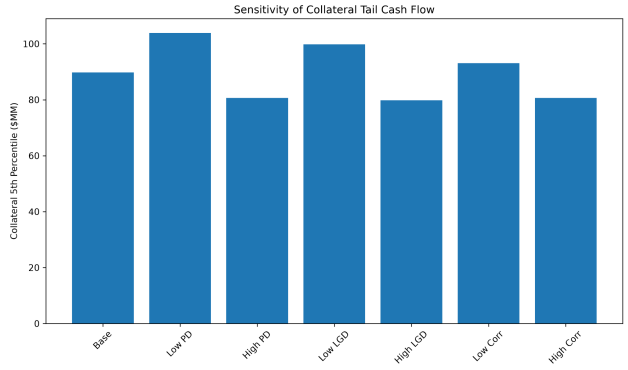
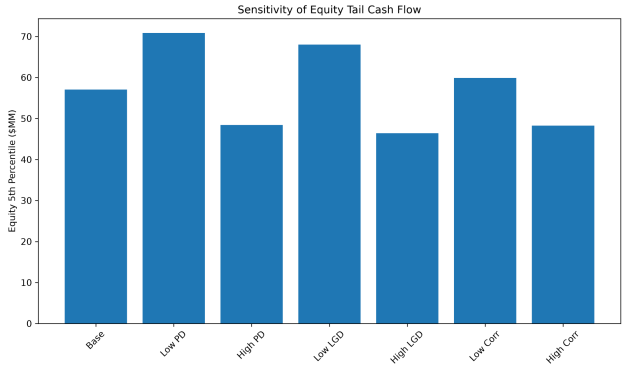


Figure 6. Equity Tail Sensitivity



Fifth-percentile cash flow by scenario. Higher default correlation leaves the mean nearly unchanged but materially worsens the lower tail.

Conclusion

The base-case simulation produces mean five-year collateral cash flow of $115.94MM, with mean Class A, Class B, and equity cash flows of $21.65MM, $11.99MM, and $82.31MM, respectively. No Class A or Class B interest or principal shortfall was observed in the 1,000 base paths under the stated waterfall assumptions. Sensitivity results show that PD primarily shifts expected default losses, LGD changes loss severity, and correlation mainly amplifies joint downside outcomes. The model therefore provides a reproducible Part 1 framework for evaluating CDO cash-flow risk before discounted valuation is introduced.